

Splunk Log Analysis Report — OVAPI

BY Ismail TP

Investigation Title

Security Investigation of OVAPI Application Logs Using Splunk Enterprise

1. Objective

The purpose of this investigation was to analyze OVAPI logs for:

- Credential exposure
- Suspicious user activity
- User deletion actions
- Possible brute-force indicators
- Sensitive information leakage

2. Environment Details

Component	Value
SIEM Platform	Splunk Enterprise
Target Application	OVAPI
Log Source	/home/ovapi/.pm2/logs/ovapi-out.log
Web Logs	/var/log/nginx/access.log
Sourcetypes	ovapi_access, access-too_small, error-too_small

3. Investigation Methodology

The investigation was conducted using:

- Username enumeration
- Password extraction from logs
- API request analysis
- DELETE request investigation

- Log correlation
- Authentication activity review

4. Splunk Queries Used

4.1 Password Extraction Query

```
index=* sourcetype=ovapi_access
("Dhamu" OR "Carlos" OR "Ismail" OR "TRIP4ZY" OR "ronaldo12" OR "subsystem6447" OR "user999" OR "Akshaylal")
| rex field=_raw "(?i)(password|passwd|pwd)[\"=: ]+(?<password>[^\s\",&])+"
| search password!=""
| table _time password _raw
```

5. Findings

5.1 Plaintext Password Exposure

The investigation identified multiple plaintext passwords stored directly in application logs.

Exposed Credentials

Username	Password
Ismail	123456
dhamu	Pass@123
carlos	pass@123
TRIP4ZY	trip4zy@123
subsystem6447	subsystem6447
user999	pass999
ronaldo12	ronaldo@321
akshaylal	offenso1005

Evidence

Figure 3 — Plaintext password exposure for Ismail.

```
> 3/13/26      {
10:12:20.000 AM  _id: new ObjectId("69b0ee1fe2fc9869af1e4bfe"),
                  username: 'ismail',
                  password: '123456',
                  secretNote: 'hy ',
                  Show all 22 lines
                  host = ovapi | source = /home/ovapi/.pm2/logs/ovapi-out.log | sourcetype = ovapi_access
```

Figure 4 — Plaintext password exposure for dhamu.

```
> 3/13/26      ... 2 lines omitted ...
12:18:19.000 PM  _id: new ObjectId("698096ece6fd328a85d90215"),
                  username: 'dhamu',
                  password: 'Pass@123',
                  ... 22 lines omitted ...
                  username: 'dhamu',
                  ... 22 lines omitted ...
                  username: 'dhamu',
                  Show all 70 lines
                  host = ovapi | source = /home/ovapi/.pm2/logs/ovapi-out.log | sourcetype = ovapi_access
```

Figure 5 — Plaintext password exposure for Carlos.

```
> 12/15/25      eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQiOiI2OTMxMjlnBjZDlkZDZmZW4NGUxZTMiLCJ1c2VybmFtZSI6ImNhcmxvYyIsIm1zQWRtaW4
4:19:19.000 PM  {
                  _id: new ObjectId("693129b60cd9dd2fec84e1e3"),
                  username: 'carlos',
                  password: 'pass@123',
                  Show all 67 lines
                  host = ovapi | source = /home/ovapi/.pm2/logs/ovapi-out.log | sourcetype = ovapi_access
```

Figure 6 — Plaintext password exposure for TRIP4ZY.

```
> 3/13/26      ... 26 lines omitted ...
10:49:34.000 AM  username: 'TRIP4ZY',
                  password: 'trip4zy@123',
                  ... 22 lines omitted ...
                  username: 'TRIP4ZY',
                  ... 45 lines omitted ...
                  username: 'TRIP4ZY',
                  password: 'trip4zy@123',
                  Show all 186 lines
                  host = ovapi | source = /home/ovapi/.pm2/logs/ovapi-out.log | sourcetype = ovapi_access
```

Figure 7 — Plaintext password exposure for subsystem6447.

```
> 5/23/26      ... 2 lines omitted ...
1:30:21.000 PM  _id: new ObjectId("6a1009d7f10846760c96d4dd"),
                  username: 'subsystem6447',
                  password: 'subsystem6447',
                  ... 21 lines omitted ...
                  username: 'subsystem6447',
                  password: 'subsystem6447',
                  Show all 46 lines
                  host = ovapi | source = /home/ovapi/.pm2/logs/ovapi-out.log | sourcetype = ovapi_access
```

Figure 8 — Plaintext password exposure for user999.

```

> 5/21/26      ... 3 lines omitted ...
12:43:23.000 PM  _id: new ObjectId("6a0eaf37f10846760c96d0e9"),
                  username: 'user999',
                  password: 'pass999',
                  ... 21 lines omitted ...
                  username: 'user999',
                  password: 'pass999',
                  Show all 47 lines
host = ovapi | source = /home/ovapi/pm2/logs/ovapi-out.log | sourcetype = ovapi_access

```

Figure 9 — Plaintext password exposure for akshaylal.

```

> 5/12/26      ... 2 lines omitted ...
11:18:21.000 PM  _id: new ObjectId("69fae416b11057c05f88b3e9"),
                  username: 'akshaylal',
                  password: 'offenso1005',
                  ... 21 lines omitted ...
                  username: 'akshaylal',
                  password: 'offenso1005',
                  Show all 46 lines
host = ovapi | source = /home/ovapi/pm2/logs/ovapi-out.log | sourcetype = ovapi_access

```

Figure 10 — Plaintext password exposure for ronaldo12 .

```

host = ovapi | source = /home/ovapi/pm2/logs/ovapi-out.log | sourcetype = ovapi_access
> 5/25/26      {
4:20:41.000 PM  _id: new ObjectId("6a141c886ca116a330c2120d"),
                  username: 'ronaldo12',
                  password: 'ronaldo@321',
                  secretNote: '<html>\r',
                  Show all 22 lines
host = ovapi | source = /home/ovapi/pm2/logs/ovapi-out.log | sourcetype = ovapi_access

```

Security Impact

This issue can lead to:

- Credential theft
- Account compromise
- Privilege escalation
- Unauthorized access
- Lateral movement

Severity

Critical

5.2 User Deletion Activity

The investigation identified multiple DELETE API requests targeting user accounts.

Observed DELETE Requests

```

DELETE /api/user/carlos
DELETE /api/user/dhamu
DELETE /api/user/ismail
DELETE /api/user/TRIP4ZY

```

Evidence

Figure 10 — DELETE request targeting Carlos account.

>	12/18/25	10:42:17.000 AM	10.8.0.18 - - [18/Dec/2025:05:12:17 +0000] "DELETE /api/user/carlos HTTP/1.1" 200 98 "-" "PostmanRuntime/7.49.1"
			host = ovapi : source = /var/log/nginx/access.log : sourcetype = access-too_small
>	12/18/25	10:42:17.000 AM	2025/12/18 05:12:17 [error] 45331#45331: *1347831 connect() failed (111: Connection refused) while connecting to upstream, client: 10.8.0.18, server: ovapi.offenso.local, request: "DELETE /api/user/carlos HTTP/1.1", upstream: "http://[::1]:3000/api/user/carlos", host: "ovapi.offenso.local"
			host = ovapi : source = /var/log/nginx/error.log : sourcetype = error-too_small

Figure 11 — DELETE request targeting dhamu account.

i	Time	Event
>	4/10/26	192.168.56.102 - - [10/Apr/2026:05:29:10 +0000] "DELETE /api/user/dhamu HTTP/1.1" 200 98 "https://ovapi.offenso.local/user/abilesh" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
	10:59:10.000 AM	host = ovapi : source = /var/log/nginx/access.log : sourcetype = access-too_small

Figure 12 — DELETE request targeting Ismail account.

host"ovapi" "Ismail" "delete"

Time range: All time

Q

7 events (before 5/26/26 10:57:46.000 AM)

No Event Sampling

Job

||

Verbose Mode

Events (7)

Patterns

Statistics

Visualization

Timeline format

Zoom Out

Zoom to Selection

X Deselect

1 day per column

Format

Show: 50 Per Page

View: List

Hide Fields

All Fields

SELECTED FIELDS

host 1

source 1

sourcetype 1

INTERESTING FIELDS

date_hour 1

date_mday 3

date_minute 4

date_month 2

date_second 5

date_wday 3

date_year 1

date_zone 1

index 1

linecount 1

punct 4

splunk_server 1

timeendpos 3

timestartpos 3

i	Time	Event
>	4/10/26 10:59:11.000 AM	192.168.56.102 - - [10/Apr/2026:05:29:11 +0000] "DELETE /api/user/ismail%201000 HTTP/1.1" 200 98 "https://ovapi.offenso.local/user/abilesh" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" host = ovapi : source = /var/log/nginx/access.log : sourcetype = access-too_small
>	4/10/26 10:59:11.000 AM	192.168.56.102 - - [10/Apr/2026:05:29:11 +0000] "DELETE /api/user/ismail%2010 HTTP/1.1" 200 98 "https://ovapi.offenso.local/user/abilesh" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" host = ovapi : source = /var/log/nginx/access.log : sourcetype = access-too_small
>	4/10/26 10:59:10.000 AM	192.168.56.102 - - [10/Apr/2026:05:29:10 +0000] "DELETE /api/user/ismail HTTP/1.1" 200 98 "https://ovapi.offenso.local/user/abilesh" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" host = ovapi : source = /var/log/nginx/access.log : sourcetype = access-too_small
>	4/10/26 10:54:17.000 AM	192.168.56.102 - - [10/Apr/2026:05:24:17 +0000] "DELETE /api/user/ismail%201 HTTP/1.1" 200 50 "https://ovapi.offenso.local/user/abilesh" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" host = ovapi : source = /var/log/nginx/access.log : sourcetype = access-too_small
>	4/10/26 10:54:17.000 AM	192.168.56.102 - - [10/Apr/2026:05:24:17 +0000] "DELETE /api/user/ismail%201 HTTP/1.1" 200 50 "https://ovapi.offenso.local/user/abilesh" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" host = ovapi : source = /var/log/nginx/access.log : sourcetype = access-too_small
>	3/16/26 10:30:21.000 AM	172.16.0.202 - - [16/Mar/2026:05:00:21 +0000] "DELETE /api/user/ismail%2011 HTTP/1.1" 200 98 "-" "PostmanRuntime/7.52.0" host = ovapi : source = /var/log/nginx/access.log : sourcetype = access-too_small

Figure 13 — DELETE request targeting TRIP4ZY account.

>	4/10/26	192.168.56.102 - - [10/Apr/2026:05:29:10 +0000] "DELETE /api/user/TRIP4ZY HTTP/1.1" 200 98 "https://ovapi.offenso.local/user/abilesh" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
	10:59:10.000 AM	host = ovapi : source = /var/log/nginx/access.log : sourcetype = access-too_small

Observations

The logs indicate:

- User deletion functionality exposed through API endpoints
- Requests performed from internal IP addresses
- Requests made through browsers and Postman
- Successful HTTP 200 responses for deletion operations

Security Impact

Potential risks include:

- Unauthorized user deletion

- Administrative abuse
- Account destruction
- Insider threat activity

Severity

High

5.3 JWT Token Exposure

JWT tokens were identified inside application logs.

Evidence

Figure 14 — JWT token exposure inside OVAPI logs.

[Insert Screenshot Here]

Security Impact

JWT exposure may lead to:

- Session hijacking
- Token replay attacks
- Unauthorized authentication
- Privilege escalation

Severity

Critical

6. Brute Force Analysis

The logs were reviewed for:

- Repeated failed logins
- Password spraying
- Authentication failures
- High-volume login attempts

Result

>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:54.372295+00:00	ovwa	sshd[25221]: Failed password for invalid user	offenso from 10.8.2.181 port 37176 ssh2
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:54.183544+00:00	ovwa	sshd[25217]: Failed password for invalid user	offenso from 10.8.2.181 port 37144 ssh2
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:54.152207+00:00	ovwa	sshd[25209]: Disconnecting invalid user	offenso 10.8.2.181 port 41958: Too many authentication failures [preauth]
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:54.151786+00:00	ovwa	sshd[25209]: error: maximum authentication attempts exceeded for invalid user	offenso from 10.8.2.181 port 41958 ssh2 [preauth]
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:53.534414+00:00	ovwa	sshd[25211]: Failed password for invalid user	offenso from 10.8.2.181 port 37074 ssh2
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:53.531529+00:00	ovwa	sshd[25209]: Failed password for invalid user	offenso from 10.8.2.181 port 41958 ssh2
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:53.494574+00:00	ovwa	sshd[25219]: Failed password for invalid user	offenso from 10.8.2.181 port 37152 ssh2
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:53.457924+00:00	ovwa	sshd[25227]: Failed password for invalid user	offenso from 10.8.2.181 port 35750 ssh2
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:53.374564+00:00	ovwa	sshd[25207]: Disconnecting invalid user	offenso 10.8.2.181 port 41924: Too many authentication failures [preauth]
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:53.374231+00:00	ovwa	sshd[25207]: error: maximum authentication attempts exceeded for invalid user	offenso from 10.8.2.181 port 41924 ssh2 [preauth]
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:53.370590+00:00	ovwa	sshd[25201]: Disconnecting invalid user	offenso 10.8.2.181 port 41888: Too many authentication failures [preauth]
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:53.370290+00:00	ovwa	sshd[25201]: error: maximum authentication attempts exceeded for invalid user	offenso from 10.8.2.181 port 41888 ssh2 [preauth]
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:53.360239+00:00	ovwa	sshd[25205]: Disconnecting invalid user	offenso 10.8.2.181 port 41912: Too many authentication failures [preauth]
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:53.359875+00:00	ovwa	sshd[25205]: error: maximum authentication attempts exceeded for invalid user	offenso from 10.8.2.181 port 41912 ssh2 [preauth]
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:53.321571+00:00	ovwa	sshd[25215]: Failed password for invalid user	offenso from 10.8.2.181 port 37138 ssh2
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:53.277121+00:00	ovwa	sshd[25213]: Failed password for invalid user	offenso from 10.8.2.181 port 37128 ssh2
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:53.245716+00:00	ovwa	sshd[25203]: Disconnecting invalid user	offenso 10.8.2.181 port 41910: Too many authentication failures [preauth]
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:53.245542+00:00	ovwa	sshd[25203]: error: maximum authentication attempts exceeded for invalid user	offenso from 10.8.2.181 port 41910 ssh2 [preauth]
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:52.826590+00:00	ovwa	sshd[25225]: Failed password for invalid user	offenso from 10.8.2.181 port 35740 ssh2
>	5/12/26 7:45:00.000 PM	2026-05-12T14:17:52.793448+00:00	ovwa	sshd[25223]: Failed password for invalid user	offenso from 10.8.2.181 port 35738 ssh2

Observed Activity

The logs primarily contained:

- Successful authenticated requests
- Plaintext credential storage
- DELETE API activity
- JWT exposure

Conclusion

The incident appears related to:

- Insecure logging practices
- Sensitive data exposure
- Authenticated destructive actions

rather than brute-force attacks.

7. Security Issues Identified

Severity	Issue
Critical	Plaintext password logging
Critical	JWT token exposure
High	User deletion API abuse
High	Sensitive debug logging
Medium	Excessive information disclosure

8. Recommendations

Immediate Actions

1. Remove password logging immediately
2. Rotate all exposed credentials
3. Invalidate exposed JWT tokens
4. Restrict DELETE API access
5. Review user access permissions
6. Implement audit logging

Long-Term Improvements

- Mask sensitive fields in logs
- Use secure logging standards
- Implement RBAC controls
- Monitor DELETE operations
- Add SIEM alerting rules
- Enforce MFA

9. Final Conclusion

The investigation identified:

- Multiple plaintext credentials stored in logs
- JWT token exposure
- User deletion API activity
- Sensitive information leakage

No conclusive brute-force activity was observed during this investigation.

The primary security concern is insecure application logging combined with exposed destructive API functionality.